COMPUTER VISION & WASTE CLASSIFICATION

TrashNet Transfer Learning with MobileNet

Automated waste categorization pipeline built with TensorFlow and Keras

Environment & Setup

TensorFlow 2.21.0

Core deep learning framework initialized.

CPU Environment

Standard compute backend running training.

MobileNet Architecture

Lightweight backbone for image feature extraction.

TrashNet Dataset Architecture

2,527 Total Images

Standard resized benchmark images.

80 / 20 Split

2,022 training files and 505 validation files.

6 Waste Categories

Cardboard, glass, metal, paper, plastic, trash.

Transfer Learning Design

Frozen Base

Pre-trained MobileNet weights frozen to retain features.

Global Average Pooling

Spatial dimensions reduced efficiently.

Softmax Dense Head

6-class output layer mapping waste categories.

Training Results

82.77%

Validation Accuracy achieved after 5 epochs

Questions & Discussion

Thank you for your attention